

8

Multiple state models

8.1 Summary

In this chapter we reformulate the survival model introduced in Chapter 2 as an example of a multiple state model. We then introduce several other multiple state models which are useful for different types of life insurance policies, for example when benefits depend on the policyholder's health as well as survival, or when extra benefits are payable where death is accidental.

A general definition of a multiple state model, together with assumptions and notation, is given in Section 8.3. In Section 8.4 we discuss the derivation of formulae for probabilities and in Section 8.5 we consider the numerical evaluation of these probabilities. We then extend ideas from earlier chapters, first by introducing functions to value annuity and insurance benefits in Section 8.6, then we consider premium calculation in Section 8.7 and policy values in Section 8.8.

The general results in this chapter are derived for continuous-time transitions, for example, from healthy to disabled. In Section 8.10 we consider the discrete time model, where the states are observed at discrete intervals.

In Section 8.9 we show how multiple state models can be applied to specific long-term health contracts, including disability income insurance, critical illness insurance and funding of continuing care retirement communities.

8.2 Examples of multiple state models

Multiple state models are one of the most exciting developments in actuarial science in recent years. They are a natural tool for many important areas of practical interest to actuaries. They are intuitive, and easy to work with using some straightforward but powerful numerical techniques. They also simplify and provide a sound foundation for pricing and valuing complex insurance contracts. In this section we illustrate some of the applications of multiple state models, using four examples which are common in current actuarial practice,

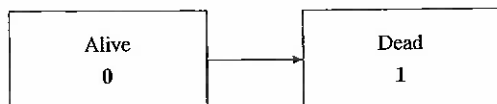


Figure 8.1 The alive-dead model.

and which offer some insight into the different levels of complexity that can be incorporated. In Section 8.9 we consider some further, more complex examples used in long-term health insurance and related contracts.

8.2.1 The alive-dead model

So far, we have modelled the uncertainty over the duration of an individual's future lifetime by regarding the future lifetime as a random variable, T_x , for an individual currently aged x . The cumulative distribution function of T_x is $\Pr[T_x \leq t] = F_x(t) = {}_tq_x$, and the survival function is $\Pr[T_x > t] = S_x(t) = {}_tp_x$. This is a **probabilistic model** in the sense that for an individual aged x we have a single random variable, T_x , whose distribution is assumed to be known, and for which all associated probabilities can be determined.

We can represent this model diagrammatically as shown in Figure 8.1. Our individual is, at any time, in one of two states, 'Alive' and 'Dead'. For convenience we label these states '0' and '1', respectively. Transition from State 0 to State 1 is allowed, as indicated by the direction of the arrow, but transitions in the opposite direction cannot occur. This is an example of a **multiple state model** with two states.

We can use this multiple state model to reformulate our survival model as follows. Suppose we have a life aged $x \geq 0$ at time $t = 0$. For each $t \geq 0$, we define a random variable $Y(t)$ which takes one of the two values 0 and 1. The event ' $Y(t) = 0$ ' means that our individual is alive at age $x + t$, and ' $Y(t) = 1$ ' means that our individual died before age $x + t$. The set of random variables $\{Y(t)\}_{t \geq 0}$ is an example of a **continuous time stochastic process**. A continuous time stochastic process is a collection of random variables indexed by a continuous time variable. For all t , $Y(t)$ is either 0 or 1, and T_x is connected to this model as the time at which $Y(t)$ jumps from 0 to 1, that is

$$T_x = \max\{t : Y(t) = 0\}.$$

The alive-dead model represented by Figure 8.1 captures all the life contingent information that is necessary for calculating insurance premiums and policy values for policies where payments (premiums, benefits and expenses) depend only on whether the individual is alive or dead at any given age. We have seen examples of these contracts, such as term insurance or deferred annuities,

in previous chapters. But there are more complicated forms of insurance which require more sophisticated models. We introduce some examples in this section, which we will use to illustrate the development of functions for calculating premiums and policy values. Further examples are presented in Section 8.9. Each of these models consists of a finite set of states with arrows indicating possible movements between some, but not necessarily all, pairs of states. Each state represents the status of an individual policy. Loosely speaking, each model is appropriate for a given insurance policy, in the sense that the condition for a payment relating to the policy – for example a premium, an annuity payment or a sum insured – is either that the individual is in a specified state at that time or that the individual makes an instantaneous transfer between a specified pair of states at that time.

8.2.2 Term insurance with increased benefit on accidental death

Suppose we are interested in a term insurance policy under which the death benefit is \$100 000 if death is due to an accident during the policy term and \$50 000 if it is due to any other cause. The alive–dead model in Figure 8.1 is not sufficient for this policy since, when the individual dies – that is, transfers from State 0 to State 1 – we do not know whether death was due to an accident, and so we do not know the amount of the death benefit to be paid.

An appropriate model for this policy is shown in Figure 8.2. This model has three states, and we can define a continuous time stochastic process, $\{Y(t)\}_{t \geq 0}$, where each random variable $Y(t)$ takes one of the values 0, 1 and 2. Hence, for example, the event ' $Y(t) = 1$ ' indicates that the individual, who was aged x at time $t = 0$, has died from an accident before age $x + t$.

The model in Figure 8.2 is an extension of the model in Figure 8.1. In both cases an individual starts by being alive, that is, starts in State 0, and, at some future time, dies. The difference is that we now distinguish between deaths due to accident and deaths due to other causes. Notice that it is the benefits provided

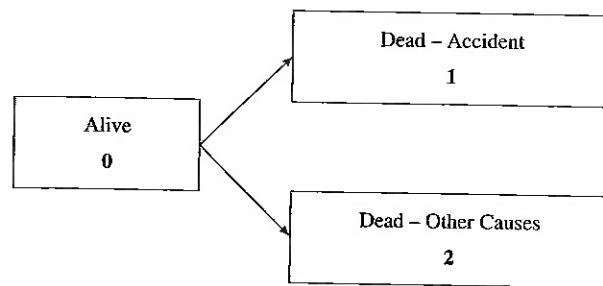


Figure 8.2 The accidental death model.

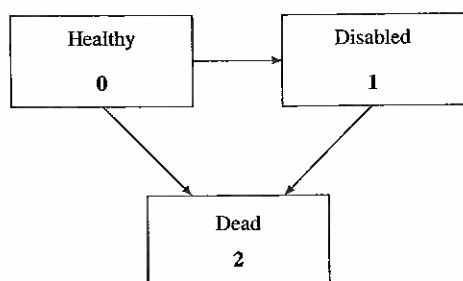


Figure 8.3 The permanent disability model.

by the insurance policy that determine the design of the model. Because the benefit is contingent on the cause of death, the model must specify the cause of death appropriately.

8.2.3 The permanent disability model

Figure 8.3 shows a model appropriate for a policy which provides some or all of the following payments:

- (i) an annuity while permanently disabled,
- (ii) a lump sum on becoming permanently disabled,
- (iii) a lump sum on death, and
- (iv) premiums payable while healthy.

An important feature of this model is that disability is permanent – there is no arrow from State 1 back to State 0.

8.2.4 The sickness–death model

In Chapter 1 we described disability income insurance, which pays an income benefit during periods of sickness. Figure 8.4 shows the sickness–death model which can be used for a policy which provides an annuity during periods of sickness, with premiums payable while the person is healthy. It could also be used for valuing lump sum payments contingent on becoming sick or on dying. The model represented by Figure 8.4 differs from that in Figure 8.3 in only one respect: it is possible to transfer from State 1 to State 0, that is, to recover from an illness.

This model illustrates an important general feature of multiple state models which was not present for the models in Figures 8.1, 8.2 and 8.3. This feature is the possibility of entering one or more states many times. In terms of our interpretation of the model, this means that several periods of sickness could occur before death, with healthy (premium paying) periods in between.

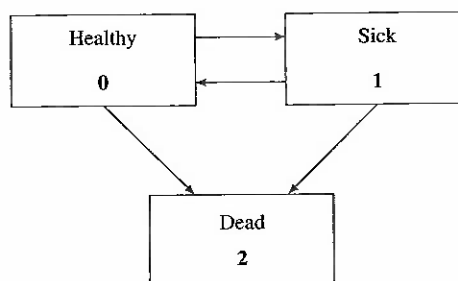


Figure 8.4 The sickness-death model for disability income insurance.

Tables of probabilities and actuarial functions for a parameterization of this model that we call the Standard Sickness-Death Model are given in Appendix D.

8.3 Assumptions and notation

The multiple state models introduced above are all extremely useful in an insurance context. We study several of these models in detail later in this chapter. Before doing so, we need to introduce some assumptions and some notation.

In this section we consider a general multiple state model. We have a finite set of $m + 1$ states labelled $0, 1, \dots, m$, with instantaneous transitions being possible between selected pairs of states. These states represent different conditions for an individual. For each $t \geq 0$, the random variable $Y(t)$ takes one of the values $0, 1, \dots, m$, and we interpret the event $Y(t) = i$ to mean that the individual is in State i at age $x + t$. The set of random variables $\{Y(t)\}_{t \geq 0}$ is then a continuous time stochastic process.

The multiple state model will be an appropriate model for an insurance policy if the payment of benefits or premiums is dependent on the life being in a given state or moving between a given pair of states, as illustrated in the examples in the previous section. Note that in these examples there is a natural starting state for the policy, which we always label State 0. This is the case for all insurance examples based on multiple state models. For example, a policy providing an annuity during periods of sickness in return for premiums payable while healthy, as described in Section 8.2.4 and illustrated in Figure 8.4, would be issued to a person who was healthy at the start of the policy.

Assumption 8.1 We assume that for any states i and j and any times t and $t + s$, where $s \geq 0$, the conditional probability $\Pr[Y(t + s) = j \mid Y(t) = i]$ is well defined, in the sense that its value does not depend on any information about the process before time t .

Intuitively, this means that the probabilities of future events for the process are completely determined by knowing the current state of the process. In particular, these probabilities do not depend on how the process arrived at the current state or how long it has been in the current state. This property, that probabilities of future events depend on the present but not on the past, is known as the **Markov property**. Using the language of probability theory, we are assuming that $\{Y(t)\}_{t \geq 0}$ is a Markov process.

Assumption 8.1 was not made explicitly for the models represented by Figures 8.1 and 8.2 since it was unnecessary, given our interpretation of these models. In each of these two cases, if we know that the process is in State 0 at time x (so that the individual is alive at age x) then we know the past of the process (the individual was alive at all ages before x). Assumption 8.1 is more interesting in relation to the models in Figures 8.3 and 8.4. Suppose, for example, in the sickness-death model (Figure 8.4) we know that $Y(t) = 1$, which means that the individual is sick at time t . Then Assumption 8.1 says that the probability of any future move after time t , either recovery or death, does not depend on any further information, such as how long the life has been sick up to time t , or how many different periods of sickness the life has experienced up to time t . In practice, we might believe that the probability of recovery in, say, the next week would depend on how long the current sickness has already lasted. If the current sickness has already lasted for, say, six months then it is likely to be a very serious illness and recovery within the next week is possible but not likely; if the current sickness has lasted only one day so far, then it may well be a trivial illness and recovery within a week could be very likely. It is important to understand the limitations of any model and also to bear in mind that no model is a perfect representation of reality. Assumption 8.1 can be relaxed to allow for some dependency on the process history, but these more general (non-Markov) models are beyond the scope of this book.

Assumption 8.2 We assume that for any positive interval of time h ,

$$\Pr[2 \text{ or more transitions within a time period of length } h] = o(h).$$

Recall that any function of h , say $g(h)$, is said to be $o(h)$ if

$$\lim_{h \rightarrow 0} \frac{g(h)}{h} = 0.$$

Intuitively, a function is $o(h)$ if, as h converges to 0, the function converges to zero faster than h . So, for example, h^k is $o(h)$ if and only if $k > 1$.

Assumption 8.2 tells us that for a very small time interval t to $t + h$, the probability of two or more transitions is vanishingly small. This assumption is unnecessary for the models in Figures 8.1 and 8.2 since in both cases only one transition can ever take place. However, it is an assumption we need to make

for technical reasons for the models in Figures 8.3 and 8.4. In these cases, given our interpretation of the models, it is not unreasonable.

In Chapter 2 we introduced the standard actuarial notation for what we are now calling the alive-dead model, as shown in Figure 8.1; specifically, ${}_t p_x$, ${}_t q_x$, and μ_x . For multiple state models with more than two states, or more than one possible transition, we need a more flexible notation. We introduce the following notation for a general multiple state model.

Notation For states i and j in a multiple state model and for $x, t \geq 0$, we define

$${}_t p_x^{ij} = \Pr[Y(x+t) = j \mid Y(x) = i], \quad (8.1)$$

$${}_t \bar{p}_x^{ii} = \Pr[Y(x+s) = i \text{ for all } s \in [0, t] \mid Y(x) = i], \quad (8.2)$$

so that ${}_t p_x^{ij}$ is the probability that a life currently aged x and currently in State i is in State j at age $x+t$, where j may be equal to i , while ${}_t \bar{p}_x^{ii}$ is the probability that a life currently aged x and currently in State i stays in State i continuously throughout the period from age x to age $x+t$.

We define

$$\mu_x^{ij} = \lim_{h \rightarrow 0^+} \frac{{}_h p_x^{ij}}{h} \quad \text{for } i \neq j. \quad (8.3)$$

Assumption 8.3 For all states i and j and all ages $x \geq 0$, we assume that ${}_t p_x^{ij}$ is a differentiable function of t .

Assumption 8.3 is a technical assumption needed to ensure that the mathematics proceeds smoothly. The first consequence of this assumption is that the limit in (8.3) always exists, which means that μ_x^{ij} always exists (though it may be zero for impossible transitions). Also, the probability of a transition taking place in a time interval of length h converges to 0 as h converges to 0. We also assume that μ_x^{ij} is a bounded and integrable function of x . These assumptions are not too restrictive in practice. However, there are some circumstances where we need to put aside Assumption 8.3, for example, when transitions between states occur at specific ages. This is discussed further in Chapter 9. To help interpret the multiple state model notation, we start by comparing it with standard actuarial notation, in terms of the alive-dead model (Figure 8.1) where we may use either notation. We make the following observations:

- For the alive-dead model, ${}_t p_x^{00}$ is the same as ${}_t p_x$, and ${}_t p_x^{01}$ is the same as ${}_t q_x$.
- ${}_t p_x^{10} = 0$ since backward transitions, 'Dead' to 'Alive', are not permitted in this model.
- μ_x^{01} is the same as μ_x , the force of mortality at age x .

In the general case, with states $0, 1, 2, \dots, m$, we refer to μ_x^{ij} as the **force of transition** or **transition intensity** between states i and j at age x .

An **absorbing state** is a state from which no exit is possible. For example, in the sickness-death model, State 2 (dead) is an absorbing state, but the other two states are not. If State i is an absorbing state, then $\mu_x^{ij} = {}_t p_x^{ij} = 0$ for all $j \neq i$.

Another way of expressing formula (8.3) is to write for $h > 0$

$${}_h p_x^{ij} = h \mu_x^{ij} + o(h) \quad \text{for } i \neq j. \quad (8.4)$$

From this formulation we can say that, for small positive values of h ,

$${}_h p_x^{ij} \approx h \mu_x^{ij} \quad \text{for } i \neq j. \quad (8.5)$$

This is equivalent to formula (2.8) in Chapter 2 for the alive-dead model and will be very useful to us.

Example 8.1 Explain why, for a general multiple state model, ${}_t \bar{p}_x^{ii}$ is not equivalent to ${}_t p_x^{ii}$. Write down an inequality linking these two probabilities and explain why

$${}_t \bar{p}_x^{ii} = {}_t p_x^{ii} + o(t). \quad (8.6)$$

Solution 8.1 From formulae (8.1) and (8.2) we can see that ${}_t \bar{p}_x^{ii}$ is the probability that the process/individual does not leave State i between ages x and $x + t$, whereas ${}_t p_x^{ii}$ is the probability that the process/individual is in State i at age $x + t$, in both cases given that the process was in State i at age x . The important distinction is that ${}_t p_x^{ii}$ includes the possibility that the process leaves State i between ages x and $x + t$, provided it is back in State i at age $x + t$. For any individual state which either (a) can never be left or (b) can never be re-entered once it has been left, these two probabilities are equivalent. This applies to all the states in the models illustrated in Figures 8.1, 8.2, 8.3, and 8.4 except States 0 and 1 in Figure 8.4.

The following inequality is always true since the left-hand side is the probability of an event which is included in the set of events whose probability is on the right-hand side

$${}_t \bar{p}_x^{ii} \leq {}_t p_x^{ii}.$$

The difference between these two probabilities is the probability of those paths where the process makes two or more transitions between ages x and $x + t$ so that it is back in State i at age $x + t$. From Assumption 8.2 we know that this probability is $o(t)$. This gives us formula (8.6). $\square$

Example 8.2 Show that, for a general multiple state model and for $h > 0$,

$${}_h p_x^{ii} = 1 - h \sum_{j=0, j \neq i}^m \mu_x^{ij} + o(h). \quad (8.7)$$

Solution 8.2 First note that $1 - {}_h p_x^{ii}$ is the probability that the process *does* leave State i at some time between ages x and $x + h$, possibly returning to State i before age $x + h$. If the process leaves State i between ages x and $x + h$ then at age $x + h$ it must be in some State j ($j \neq i$) or be in State i having made at least two transitions in the time interval of length h . Using formula (8.4) and Assumption 8.2, the sum of these probabilities is

$$h \sum_{j=0, j \neq i}^m \mu_x^{ij} + o(h),$$

which proves (8.7). □

Note that, combining (8.6) and (8.7) we have

$${}_h p_x^{ii} = 1 - h \sum_{j=0, j \neq i}^m \mu_x^{ij} + o(h). \quad (8.8)$$

Example 8.3 For the sickness-death model in Figure 8.4, you are given

x	$10p_x^{00}$	$10p_x^{01}$	$10p_x^{11}$	$10p_x^{10}$
40	0.93705	0.01953	0.70349	0.23942
50	0.83930	0.06557	0.81211	0.06057

- (a) Calculate (i) ${}_{20}p_{40}^{00}$, (ii) ${}_{20}p_{40}^{10}$, (iii) ${}_{20}p_{40}^{02}$.
 (b) Given that a healthy life aged 40 dies before age 60, what is the probability that she was healthy at age 50?

Solution 8.3

- (a) (i) ${}_{20}p_{40}^{00}$ is the probability that (40) is healthy in 20 years, given that she is healthy now. We need to calculate this from the information on transition probabilities from age 40 to 50 and from age 50 to 60. Given that (40) is currently healthy, to be healthy at time 20, she must either be healthy at time 10, or sick at time 10, and then healthy again at time 20. So the required probability is

$${}_{20}p_{40}^{00} = {}_{10}p_{40}^{00} {}_{10}p_{50}^{00} + {}_{10}p_{40}^{01} {}_{10}p_{50}^{10} = 0.78765.$$

- (ii) ${}_{20}p_{40}^{10}$ is the probability that (40) is healthy in 20 years, given that she is sick now. The two possibilities for (40), based on the state at time

$t = 10$ are that she was sick at time 10, and healthy at time 20, or she was healthy at time 10, and healthy at time 20. So the required probability is

$${}_{20}p_{40}^{10} = {}_{10}p_{40}^{11} {}_{10}p_{50}^{10} + {}_{10}p_{40}^{10} {}_{10}p_{50}^{00} = 0.24356.$$

- (iii) ${}_{20}p_{40}^{02}$ is the probability that a healthy life aged 40 dies within 20 years. There are three paths to consider.

Path 1: (40) dies before time 10, given that she is healthy currently. The probability of this is

$${}_{10}p_{40}^{02} = 1 - {}_{10}p_{40}^{00} - {}_{10}p_{40}^{01} = 0.04342.$$

Path 2: (40) is healthy at time 10, but dies between time 10 and time 20. The probability of this is

$${}_{10}p_{40}^{00} {}_{10}p_{50}^{02} = {}_{10}p_{40}^{00} (1 - {}_{10}p_{50}^{00} - {}_{10}p_{50}^{01}) = 0.08914.$$

Path 3: (40) is sick at time 10, and dies between time 10 and time 20. The probability of this is

$${}_{10}p_{40}^{01} {}_{10}p_{50}^{12} = {}_{10}p_{40}^{01} (1 - {}_{10}p_{50}^{10} - {}_{10}p_{50}^{11}) = 0.00249.$$

Hence

$${}_{20}p_{40}^{02} = 0.04341 + 0.08912 + 0.00248 = 0.13505.$$

- (b) We want to find $\Pr[(40) \text{ is Healthy at age 50} | \text{Dead before age 60}]$. Using Bayes' theorem for the conditional probability,

$$\begin{aligned} & \Pr[(40) \text{ is Healthy at age 50} | \text{Dead before age 60}] \\ &= \frac{\Pr[(40) \text{ is Healthy at age 50 and Dead before age 60}]}{\Pr[(40) \text{ is Dead before age 60}]} \\ &= \frac{{}_{10}p_{40}^{00} {}_{10}p_{50}^{02}}{{}_{20}p_{40}^{02}} = \frac{0.08914}{0.13505} = 0.66007. \end{aligned}$$

□

8.4 Formulae for probabilities

In this section we show how to derive formulae for all probabilities in terms of the transition intensities, which we assume to be known. This is the same approach as we adopted in Chapter 2, where we assumed the force of mortality, μ_x , was known and derived formula (2.20) for ${}_t p_x$ in terms of μ_{x+t} .

The fact that all probabilities can be expressed in terms of the transition intensities is important. It tells us that the transition intensities

$$\mu_x^{ij}; x \geq 0; i, j = 0, 1, \dots, m, i \neq j$$

are fundamental quantities which determine everything we need to know about a multiple state model.

The first result generalizes formula (2.20) from Chapter 2, and is valid for any multiple state model. It gives a formula for ${}_t p_x^{\bar{ii}}$ in terms of all the transition intensities out of State i , μ_x^{ij} .

For any State i in a multiple state model with $m + 1$ states, satisfying Assumptions 8.1 to 8.3,

$${}_t p_x^{\bar{ii}} = \exp \left\{ - \int_0^t \sum_{j=0, j \neq i}^m \mu_{x+s}^{ij} ds \right\}. \quad (8.9)$$

We can derive this as follows. For any $h > 0$, the probability ${}_{t+h} p_x^{\bar{ii}}$ is the probability that the life stays in State i throughout the time period $[0, t + h]$, given that the life was in State i at age x . We can split this event into two sub-events:

- the life stays in State i from age x until (at least) age $x + t$, given that it was in State i at age x , and then
- the life stays in State i from age $x + t$ until (at least) age $x + t + h$, given that it was in State i at age $x + t$ (note the different conditioning).

The probabilities of these two sub-events are ${}_t p_x^{\bar{ii}}$ and ${}_h p_{x+t}^{\bar{ii}}$, respectively, and, using the rules for conditional probabilities, we have

$${}_{t+h} p_x^{\bar{ii}} = {}_t p_x^{\bar{ii}} {}_h p_{x+t}^{\bar{ii}}.$$

Using equation (8.7), this can be rewritten as

$${}_{t+h} p_x^{\bar{ii}} = {}_t p_x^{\bar{ii}} \left(1 - h \sum_{j=0, j \neq i}^m \mu_{x+t}^{ij} + o(h) \right).$$

Rearranging this equation, we get

$$\frac{{}_{t+h} p_x^{\bar{ii}} - {}_t p_x^{\bar{ii}}}{h} = - {}_t p_x^{\bar{ii}} \sum_{j=0, j \neq i}^m \mu_{x+t}^{ij} + \frac{o(h)}{h}.$$

Dividing by ${}_t p_x^{\bar{ii}}$, and letting $h \rightarrow 0$ we have

$$\begin{aligned} \frac{1}{{}_t p_x^{\bar{ii}}} \left(\frac{d}{dt} {}_t p_x^{\bar{ii}} \right) &= - \sum_{j=0, j \neq i}^m \mu_{x+t}^{ij} \\ \Rightarrow \frac{d}{dt} \log {}_t p_x^{\bar{ii}} &= - \sum_{j=0, j \neq i}^m \mu_{x+t}^{ij}. \end{aligned}$$

Integrating over $(0, t)$ gives

$$\log {}_t p_x^{\bar{ii}} - \log {}_0 p_x^{\bar{ii}} = - \int_0^t \sum_{j=0, j \neq i}^m \mu_{x+s}^{ij} ds.$$

Note that ${}_0 p_x^{\bar{ii}} = 1$, so we have

$$\begin{aligned} \log {}_t p_x^{\bar{ii}} &= - \int_0^t \sum_{j=0, j \neq i}^m \mu_{x+s}^{ij} ds \\ \Rightarrow {}_t p_x^{\bar{ii}} &= \exp \left\{ - \int_0^t \sum_{j=0, j \neq i}^m \mu_{x+s}^{ij} ds \right\} \end{aligned}$$

which proves (8.9).

We comment on this result after the next example.

Example 8.4 Consider the model for permanent disability illustrated in Figure 8.3.

(a) Explain why, for $x \geq 0$ and $t, h > 0$,

$${}_{t+h} p_x^{01} = {}_t p_x^{01} {}_h p_{x+t}^{\bar{11}} + {}_t p_x^{\bar{00}} h \mu_{x+t}^{01} + o(h). \quad (8.10)$$

(b) Hence show that

$$\frac{d}{dt} \left({}_t p_x^{01} \exp \left\{ \int_0^t \mu_{x+s}^{12} ds \right\} \right) = {}_t p_x^{\bar{00}} \mu_{x+t}^{01} \exp \left\{ \int_0^t \mu_{x+s}^{12} ds \right\}, \quad (8.11)$$

and hence that for $u > 0$,

$${}_u p_x^{01} = \int_0^u {}_t p_x^{\bar{00}} \mu_{x+t}^{01} {}_{u-t} p_{x+t}^{\bar{11}} dt. \quad (8.12)$$

(c) Give a direct intuitive derivation of formula (8.12).

Solution 8.4 (a) To derive (8.10), consider a life who is healthy at age x . The left-hand side of (8.10) is the probability that this life is alive and disabled at age $x + t + h$. We can write down a formula for this probability by conditioning on which state the life was in at age $x + t$. Either:

- the life was disabled at age $x + t$ (probability ${}_t p_x^{01}$) and remained disabled between ages $x + t$ and $x + t + h$ (probability ${}_h p_{x+t}^{11}$), or
- the life was healthy at age $x + t$ (probability ${}_t p_x^{00}$) and then became disabled between ages $x + t$ and $x + t + h$ (probability ${}_h \mu_{x+t}^{01} + o(h)$).

Combining the probabilities of these events gives (8.10). Note that the probability of the life being healthy at age $x + t$, becoming disabled before age $x + t + h$ and then dying before age $x + t + h$ is $o(h)$ since this involves two transitions in a time interval of length h .

(b) Using equation (8.7), we can rewrite equation (8.10) as

$${}_{t+h} p_x^{01} = {}_t p_x^{01} (1 - h \mu_{x+t}^{12}) + {}_t p_x^{00} h \mu_{x+t}^{01} + o(h). \quad (8.13)$$

Rearranging, dividing by h and letting $h \rightarrow 0$ gives

$$\frac{d}{dt} {}_t p_x^{01} + {}_t p_x^{01} \mu_{x+t}^{12} = {}_t p_x^{00} \mu_{x+t}^{01}.$$

Multiplying all terms in this equation by $\exp \left\{ \int_0^t \mu_{x+s}^{12} ds \right\}$, we have

$$\frac{d}{dt} \left({}_t p_x^{01} \exp \left\{ \int_0^t \mu_{x+s}^{12} ds \right\} \right) = {}_t p_x^{00} \mu_{x+t}^{01} \exp \left\{ \int_0^t \mu_{x+s}^{12} ds \right\}.$$

Integrating both sides of this equation from $t = 0$ to $t = u$, and noting that ${}_0 p_x^{01} = 0$, we have

$${}_u p_x^{01} \exp \left\{ \int_0^u \mu_{x+s}^{12} ds \right\} = \int_0^u {}_t p_x^{00} \mu_{x+t}^{01} \exp \left\{ \int_0^t \mu_{x+s}^{12} ds \right\} dt.$$

Finally, dividing both sides by $\exp \left\{ \int_0^u \mu_{x+s}^{12} ds \right\}$ and noting that, using formula (8.9),

$${}_{u-t} p_{x+t}^{11} = \exp \left\{ - \int_t^u \mu_{x+s}^{12} ds \right\},$$

we have formula (8.12).

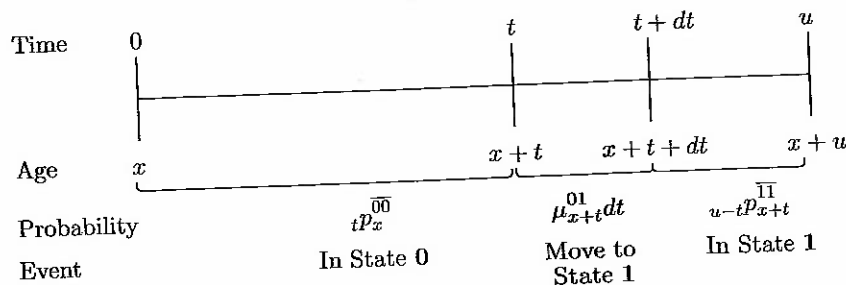


Figure 8.5 Time-line diagram for formula (8.12).

- (c) The intuitive derivation of (8.12) is as follows: for the life to move from State 0 to State 1 between ages x and $x+u$, the life must stay in State 0 until some age $x+t$, transfer to State 1 between ages $x+t$ and $x+t+dt$, where dt is small, and then stay in State 1 from age $x+t+dt$ to age $x+u$. We illustrate this event sequence using the time-line in Figure 8.5. The infinitesimal probability of this path is

$${}_tP_x^{00} \mu_{x+t}^{01} {}_{u-t}P_{x+t}^{11} dt,$$

where we have written ${}_{u-t}P_{x+t}^{11}$ instead of ${}_{u-t-dt}P_{x+t}^{11}$ since the two are equivalent for infinitesimal dt . Since the age at transfer, $x+t$, can be anywhere between x and $x+u$, the total probability, ${}_uP_x^{01}$, is the sum (i.e. integral) of these probabilities from $t=0$ to $t=u$. $\square$

We make the following comments about formula (8.9) and Example 8.4.

- (1) As we have already noted, formula (8.9) is an extension of formula (2.20), which gives the relationship between the survival probability and the force of mortality as

$${}_tP_x = \exp \left\{ - \int_0^t \mu_{x+s} ds \right\}.$$

- (2) Throughout Example 8.4 we could have replaced ${}_tP_x^{ii}$ by ${}_tP_x^{ii}$ for $i=0,1$, since for the disability insurance model, neither State 0 nor State 1 can be re-entered once it has been left.
- (3) The intuition behind formula (8.12) can be applied to other models. Although the resulting formulae are often not directly used for calculating probabilities, they can have other uses as illustrated in Exercise 8.7. For example, consider ${}_tP_x^{01}$ for the sickness-death model. We can write

$${}_t p_x^{01} = \int_0^t {}_r p_x^{00} \mu_{x+r}^{01} {}_{t-r} p_{x+r}^{\overline{11}} dr. \quad (8.14)$$

This follows because the life must spend a continuous period in State 1 immediately prior to time t if the life is in State 1 at time t ; we are integrating over all possible times of final transition to State 1 in equation (8.14).

- (4) Perhaps the most important point to note about formula (8.9) and Example 8.4 is how similar the derivations are in their basic approach. In both cases we wrote down an expression for the probability of being in the required state at age $x + t + h$ by conditioning on the state occupied at age $x + t$. This led to a formula for the derivative of the required probability which we were then able to solve. An obvious question for us is, 'Can this method be applied to a general multiple state model to derive formulae for probabilities?' The answer is, 'Yes', as we demonstrate in the following section.

8.4.1 Kolmogorov's forward equations

Let i and j be any two, not necessarily distinct, states in a multiple state model which has a total of $m + 1$ states, numbered $0, 1, \dots, m$. For $x, t, h \geq 0$, we derive the formula

$${}_{t+h} p_x^{ij} = {}_t p_x^{ij} - h \sum_{k=0, k \neq j}^m \left({}_t p_x^{ij} \mu_{x+t}^{jk} - {}_t p_x^{ik} \mu_{x+t}^{kj} \right) + o(h), \quad (8.15)$$

and hence show the main result, that

$$\frac{d}{dt} {}_t p_x^{ij} = \sum_{k=0, k \neq j}^m \left({}_t p_x^{ik} \mu_{x+t}^{kj} - {}_t p_x^{ij} \mu_{x+t}^{jk} \right). \quad (8.16)$$

Formula (8.16) gives a set of equations for a Markov process known as Kolmogorov's forward equations.

To derive Kolmogorov's forward equations, we proceed as we did in formula (8.9) and in Example 8.4. We consider the probability of being in the required state, j , at age $x + t + h$, and condition on the state at age $x + t$: either it is already in State j , in which case it needs to stay in j , or possibly leave and come back, between ages $x + t$ and $x + t + h$, or it is in some other state, say k , and must transition to State j before age $x + t + h$. Thus, we have

$${}_{t+h} p_x^{ij} = {}_t p_x^{ij} {}_h p_{x+t}^{jj} + \sum_{k=0, k \neq j}^m {}_t p_x^{ik} {}_h p_{x+t}^{kj}.$$

Using formula (8.8), we have

$${}_h p_{x+t}^{ij} = 1 - h \sum_{k=0, k \neq j}^m \mu_{x+t}^{jk} + o(h)$$

and using (8.4), we have

$${}_h p_{x+t}^{kj} = h \mu_{x+t}^{kj} + o(h).$$

So, substituting, and collecting all the $o(h)$ terms together, we have

$$\begin{aligned} {}_{t+h} p_x^{ij} &= {}_t p_x^{ij} \left(1 - h \sum_{k=0, k \neq j}^m \mu_{x+t}^{jk} \right) + \sum_{k=0, k \neq j}^m {}_t p_x^{ik} h \mu_{x+t}^{kj} + o(h) \\ \Rightarrow \frac{{}_{t+h} p_x^{ij} - {}_t p_x^{ij}}{h} &= \sum_{k=0, k \neq j}^m {}_t p_x^{ik} \mu_{x+t}^{kj} - {}_t p_x^{ij} \sum_{k=0, k \neq j}^m \mu_{x+t}^{jk} + \frac{o(h)}{h} \end{aligned}$$

and letting $h \rightarrow 0$ gives

$$\frac{d}{dt} {}_t p_x^{ij} = \sum_{k=0, k \neq j}^m \left({}_t p_x^{ik} \mu_{x+t}^{kj} - {}_t p_x^{ij} \mu_{x+t}^{jk} \right)$$

as required.

We can solve the Kolmogorov forward equations numerically, but we need boundary conditions. For any model, we use the facts that, for any state i ,

$${}_0 p_x^{ii} = 1$$

and for any two states i and j where $i \neq j$,

$${}_0 p_x^{ij} = 0.$$

8.5 Numerical evaluation of probabilities

In this section we discuss methods for the numerical evaluation of probabilities for a multiple state model given that all the transition intensities are known. In some cases, the probabilities can be calculated directly from formulae in terms of integrals, as the following example shows.

Example 8.5 Consider the permanent disability model illustrated in Figure 8.3.

(a) Suppose the transition intensities for this model are all constants, as follows

$$\mu_x^{01} = 0.0279, \quad \mu_x^{02} = 0.0229, \quad \mu_x^{12} = \mu_x^{02}.$$

Calculate ${}_{10} p_{60}^{00}$ and ${}_{10} p_{60}^{01}$.

(b) Now suppose the transition intensities for this model are as follows

$$\mu_x^{01} = a_1 + b_1 \exp\{c_1 x\}, \quad \mu_x^{02} = a_2 + b_2 \exp\{c_2 x\}, \quad \mu_x^{12} = \mu_x^{02},$$

where

$$a_1 = 4 \times 10^{-4}, \quad b_1 = 3.4674 \times 10^{-6}, \quad c_1 = 0.138155,$$

$$a_2 = 5 \times 10^{-4}, \quad b_2 = 7.5858 \times 10^{-5}, \quad c_2 = 0.087498.$$

Calculate ${}_{10}p_{60}^{00}$ and ${}_{10}p_{60}^{01}$.

Solution 8.5 For this model, neither State 0 nor State 1 can be re-entered once it has been left, so that ${}_t p_x^{ii} \equiv {}_t \bar{p}_x^{ii}$ for $i = 0, 1$ and for any $x, t \geq 0$.

(a) Using formula (8.9), we have

$${}_t p_{60}^{00} \equiv {}_t \bar{p}_{60}^{00} = \exp \left\{ - \int_0^t (0.0279 + 0.0229) ds \right\} = \exp\{-0.0508t\}, \quad (8.17)$$

giving

$${}_{10}p_{60}^{00} = \exp\{-0.508\} = 0.60170.$$

Similarly

$${}_{10-t}p_{60+t}^{11} = \exp\{-0.0229(10-t)\},$$

and we can calculate ${}_{10}p_{60}^{01}$ using formula (8.12), integrating over the time of transition from State 0 to State 1, as

$$\begin{aligned} {}_{10}p_{60}^{01} &= \int_0^{10} {}_t p_{60}^{00} \mu_{60+t}^{01} {}_{10-t}p_{60+t}^{11} dt \\ &= \int_0^{10} \exp\{-0.0508t\} \times 0.0279 \times \exp\{-0.0229(10-t)\} dt \\ &= 0.0279 \exp\{-0.229\} \int_0^{10} \exp\{-0.0279t\} dt \\ &= \exp\{-0.229\} (1 - \exp\{-0.279\}) \\ &= 0.19363. \end{aligned}$$

(b) In this case

$$\begin{aligned} {}_t p_{60}^{00} &= \exp \left\{ - \int_0^t (\mu_{60+r}^{01} + \mu_{60+r}^{02}) dr \right\} \\ &= \exp \left\{ - \left((a_1 + a_2)t + \frac{b_1}{c_1} e^{60c_1} (e^{c_1 t} - 1) + \frac{b_2}{c_2} e^{60c_2} (e^{c_2 t} - 1) \right) \right\} \end{aligned}$$

and

$${}_t p_{60}^{11} = \exp \left\{ - \int_0^t \mu_{60+r}^{12} dr \right\} = \exp \left\{ - \left(a_2 t + \frac{b_2}{c_2} e^{60c_2} (e^{c_2 t} - 1) \right) \right\}.$$

Hence ${}_{10} p_{60}^{00} = 0.58395$.

Substituting the expressions for ${}_t p_{60}^{00}$ and ${}_{10-t} p_{60+t}^{11}$ and the formula for μ_{60+t}^{01} into formula (8.12) and integrating numerically, we obtain

$${}_{10} p_{60}^{01} = 0.20577.$$

□

Probabilities of the form ${}_t p_x^{\bar{ii}}$ can be evaluated analytically provided the sum of the relevant intensities can be integrated analytically. In other cases numerical integration can be used. However, the approach used in Example 8.5 part (b) to calculate a more complicated probability, ${}_{10} p_{60}^{01}$, — that is, deriving an integral formula for the probability which can then be integrated numerically — is not tractable except in the simplest cases. Broadly speaking, this approach works if the model has relatively few states and if none of these states can be re-entered once it has been left. These conditions are met by the permanent disability model, illustrated in Figure 8.3 and used in Example 8.5, but are not met, for example, by the sickness-death model illustrated in Figure 8.4 since States 0 and 1 can both be re-entered. This means, for example, that ${}_t p_x^{01}$ is the sum of the probabilities of exactly one transition (0 to 1), plus three transitions (0 to 1, then 1 to 0, then 0 to 1 again), plus five transitions, and so on. A probability involving k transitions involves multiple integration with k nested integrals.

However, Euler's method, introduced in Chapter 7, can be used to evaluate probabilities for all models in which we are interested. The key to using this method is formula (8.15) and we illustrate it by applying it in the following example.

Example 8.6 Consider the sickness-death model illustrated in Figure 8.4. Suppose the transition intensities for this model are as follows

$$\begin{aligned}\mu_x^{01} &= a_1 + b_1 \exp\{c_1 x\}, & \mu_x^{10} &= 0.1 \mu_x^{01}, \\ \mu_x^{02} &= a_2 + b_2 \exp\{c_2 x\}, & \mu_x^{12} &= \mu_x^{02},\end{aligned}$$

where a_1, b_1, c_1, a_2, b_2 and c_2 are as in Example 8.5, part (b) (though this is a different model).

Calculate ${}_{10}p_{60}^{00}$ and ${}_{10}p_{60}^{01}$ using formula (8.15) with a step size of $h = 1/12$ years.

Solution 8.6 For this particular model, formula (8.15) gives us the two formulae

$${}_{t+h}p_{60}^{00} = {}_tp_{60}^{00} - h {}_tp_{60}^{00} (\mu_{60+t}^{01} + \mu_{60+t}^{02}) + h {}_tp_{60}^{01} \mu_{60+t}^{10} + o(h)$$

and

$${}_{t+h}p_{60}^{01} = {}_tp_{60}^{01} - h {}_tp_{60}^{01} (\mu_{60+t}^{12} + \mu_{60+t}^{10}) + h {}_tp_{60}^{00} \mu_{60+t}^{01} + o(h).$$

As in Chapter 7, we choose a small step size h , ignore the $o(h)$ terms and regard the resulting approximations as exact formulae. This procedure changes the above formulae into

$${}_{t+h}p_{60}^{00} = {}_tp_{60}^{00} - h {}_tp_{60}^{00} (\mu_{60+t}^{01} + \mu_{60+t}^{02}) + h {}_tp_{60}^{01} \mu_{60+t}^{10}$$

and

$${}_{t+h}p_{60}^{01} = {}_tp_{60}^{01} - h {}_tp_{60}^{01} (\mu_{60+t}^{12} + \mu_{60+t}^{10}) + h {}_tp_{60}^{00} \mu_{60+t}^{01}.$$

By choosing successively $t=0, h, 2h, \dots, 10-h$, we can use these formulae, together with the initial values ${}_0p_{60}^{00} = 1$ and ${}_0p_{60}^{01} = 0$, to calculate ${}_hp_{60}^{00}$, ${}_hp_{60}^{01}$, ${}_{2h}p_{60}^{00}$, ${}_{2h}p_{60}^{01}$, and so on until we have a value for ${}_{10}p_{60}^{00}$, as required. These calculations are very well suited to a spreadsheet. For a step size of $h = 1/12$ years, selected values are shown in Table 8.1. Note that the calculations have been carried out using more significant figures than are shown in this table. $\square$

The implementation of Euler's method in this example differs in two respects from the implementation in Chapter 7.

- (1) We work forward recursively from initial values for the probabilities rather than backwards from the final value of the policy value. This is determined by the boundary conditions for the differential equations.
- (2) We have two equations to solve simultaneously rather than a single equation. This is a typical feature of applying Euler's method to the calculation of probabilities for multiple state models. In general, the number of equations increases with the number of states in the model.

Table 8.1 Calculation of ${}_{10}p_{60}^{00}$ and ${}_{10}p_{60}^{01}$ using a step size $h = 1/12$ years.

t	μ_{60+t}^{01}	μ_{60+t}^{02}	μ_{60+t}^{10}	μ_{60+t}^{12}	${}_tp_{60}^{00}$	${}_tp_{60}^{01}$
0	0.01420	0.01495	0.00142	0.01495	1.00000	0.00000
$\frac{1}{12}$	0.01436	0.01506	0.00144	0.01506	0.99757	0.00118
$\frac{2}{12}$	0.01453	0.01517	0.00145	0.01517	0.99512	0.00238
$\frac{3}{12}$	0.01469	0.01527	0.00147	0.01527	0.99266	0.00358
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1	0.01625	0.01628	0.00162	0.01628	0.96977	0.01479
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$9\frac{11}{12}$	0.05473	0.03492	0.00547	0.03492	0.59189	0.20061
10	0.05535	0.03517	0.00554	0.03517	0.58756	0.20263

8.6 State-dependent insurance and annuity functions

So far we have demonstrated that multiple state models are a natural way of modelling cash flows that are state-dependent, and we have shown how to evaluate probabilities for such models given only the transition intensities between pairs of states. The next stage in our study of multiple state models is to calculate premiums and policy values for a policy represented by such a model and for these we develop actuarial functions analogous to the $\bar{A}_x$ and $\bar{a}_x$ functions of Chapters 4 and 5.

We implicitly use the indicator function approach from Section 4.6 – that is, we develop expected present values by summing the product of the probabilities, amounts and discount functions over all possible payment dates. This leads directly to intuitive formulae for the EPVs, but does not give higher moments, which are outside the scope of this book.

There is no standard actuarial notation for insurance and annuity functions in the multiple state model framework; the notation used in this chapter generalizes International Actuarial Notation, and has become widespread in actuarial practice.

State-dependent insurance benefits

Insurance benefits are conditional on making a transition into a specified state. For example, a death benefit is payable on transition into the dead state; a permanent disability insurance policy might pay a sum insured on becoming disabled.